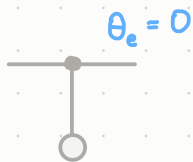


Exercise 4



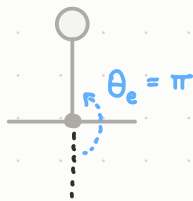
Goal Today: Time response, stability

Last week's pendulum:



$\Rightarrow$
linearized

$$A_0 = \begin{pmatrix} 0 & 1 \\ -g/l & -c/ml^2 \end{pmatrix}$$



$\Rightarrow$
linearized

$$A_\pi = \begin{pmatrix} 0 & 1 \\ +g/l & -c/ml^2 \end{pmatrix}$$

Today's goal: we read the entire future of a linear system straight from A and decide if it's stable!

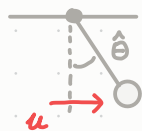
○ First we need to know what a "time response" is:

$\Rightarrow$ A time response is simply the output of a system when we apply an input, shown in the time domain:



pendulum
in
initial
condition
 $x_0 = \begin{pmatrix} \hat{\theta} \\ 0 \end{pmatrix}$

$\Rightarrow$



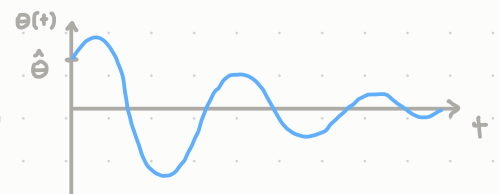
we apply
pulse input u
at $t = 0$

$\Rightarrow$



pendulum
swings

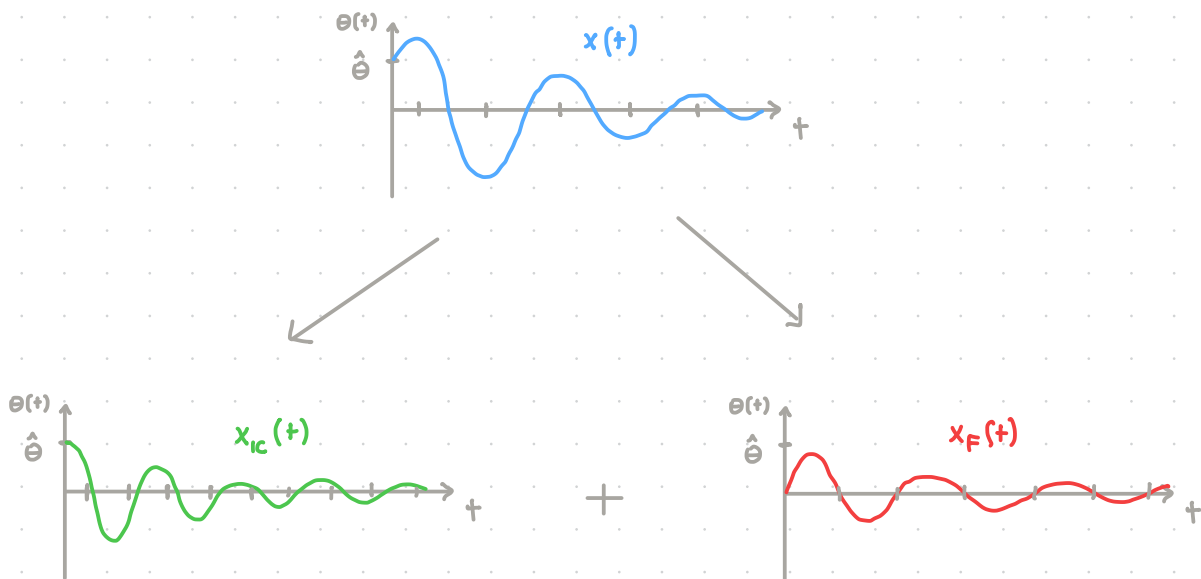
$\Rightarrow$



time response!

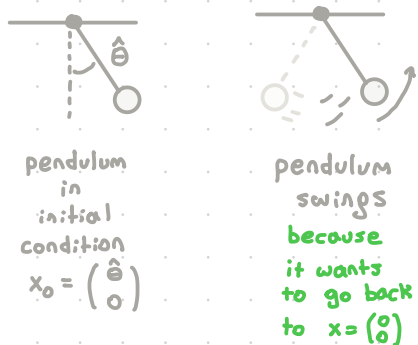
- Now, because our system is **linear** (we only look at LT systems) we can decompose our time response in two signals:

Full time response



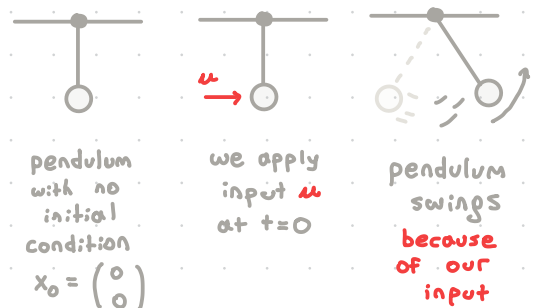
Initial Condition Response

- What if we just let the system evolve by itself from the initial condition:



Forced Response

- What if we start from initial condition $x_0 = 0$ and only applied our input:



$$x(t) = x_{ic}(t) + x_F(t)$$

$$y(t) = y_{ic}(t) + y_F(t)$$

using the property of linearity.

- Now let's look at both for a 1st order system

$$(A=a, B=b, C=c, D=d, \quad a, b, c, d \in \mathbb{R})$$

Full System

$$\begin{cases} \dot{x}(t) = a \cdot x(t) + b \cdot u(t) \\ y(t) = c \cdot x(t) + d \cdot u(t) \\ x(0) = x_0 \end{cases}$$

Initial Cond. Response

$$\begin{cases} \dot{x}(t) = a \cdot x(t) \\ y(t) = c \cdot x(t) \\ x(0) = x_0 \end{cases}$$

Forced Response

$$\begin{cases} \dot{x}(t) = a \cdot x(t) + b \cdot u(t) \\ y(t) = c \cdot x(t) + d \cdot u(t) \\ x(0) = 0 \end{cases}$$

- We know how to solve this!

$$\frac{dx}{dt} = a \cdot x$$

$$\int_{x(0)}^{x(t)} \frac{dx}{x} = \int_0^t a \cdot dt$$

$$\ln\left(\frac{x}{x_0}\right) = a \cdot t$$

$$x_{ic}(t) = e^{at} \cdot x_0$$

$$y_{ic}(t) = c \cdot e^{at} \cdot x_0$$

- The solution of this is given by:

$$x_F(t) = \int_0^t e^{a \cdot (t-\tau)} \cdot b \cdot u(\tau) d\tau$$

$$y_F(t) = c \cdot \int_0^t e^{a \cdot (t-\tau)} \cdot b \cdot u(\tau) d\tau + d \cdot u(t)$$

Full System

$$x(t) = \underbrace{e^{at} \cdot x_0}_{x_{ic}(t)} + \underbrace{\int_0^t e^{a \cdot (t-\tau)} \cdot b \cdot u(\tau) d\tau}_{x_F(t)}$$

$$y(t) = \underbrace{c \cdot e^{at} \cdot x_0}_{y_{ic}(t)} + \underbrace{c \cdot \int_0^t e^{a \cdot (t-\tau)} \cdot b \cdot u(\tau) d\tau + d \cdot u(t)}_{y_F(t)}$$

② Your Turn:

$$\begin{aligned} \dot{x}(t) &= -2x(t) + 3u(t) \\ y(t) &= x(t) \end{aligned} \quad \text{with} \quad \begin{aligned} u(t) &= 1 \\ x(0) &= 1/2 \end{aligned}$$

$y(t) = ?$ and plot y_{IC} , y_F , y . Think about the meaning of "-2"



- Of course, in general, systems are not first order and we have matrices (A, B, C, D) . But luckily, this changes **nothing** in our answers as the derivation is **exactly the same**.

$$\left\{ \begin{aligned} \dot{x}(t) &= A \cdot x(t) + B \cdot u(t) \\ y(t) &= C \cdot x(t) + D \cdot u(t) \\ x(0) &= x_0 \end{aligned} \right.$$

↙
↘

$$\left\{ \begin{aligned} x_{IC} &= e^{A^+} \cdot x_0 \\ y_{IC} &= C e^{A^+} \cdot x_0 \end{aligned} \right.$$

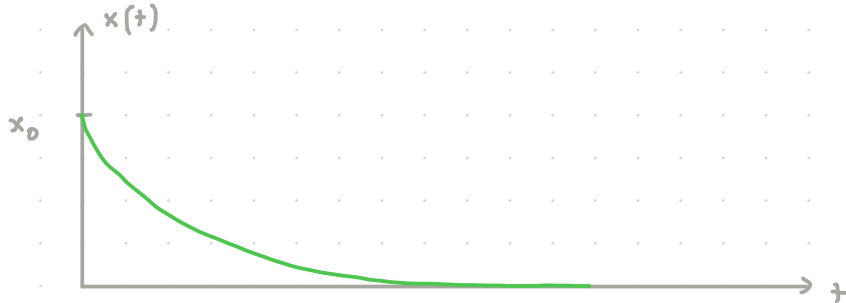
$$\left\{ \begin{aligned} x_F &= \int_0^+ e^{A(t-\tau)} B u(\tau) d\tau \\ y_F &= C \cdot \int_0^+ e^{A(t-\tau)} B u(\tau) d\tau + D u(t) \end{aligned} \right.$$

$$\left\{ \begin{aligned} x(t) &= e^{A^+} \cdot x_0 + \int_0^+ e^{A(t-\tau)} B u(\tau) d\tau \\ y(t) &= C e^{A^+} \cdot x_0 + C \cdot \int_0^+ e^{A(t-\tau)} B u(\tau) d\tau + D u(t) \end{aligned} \right.$$

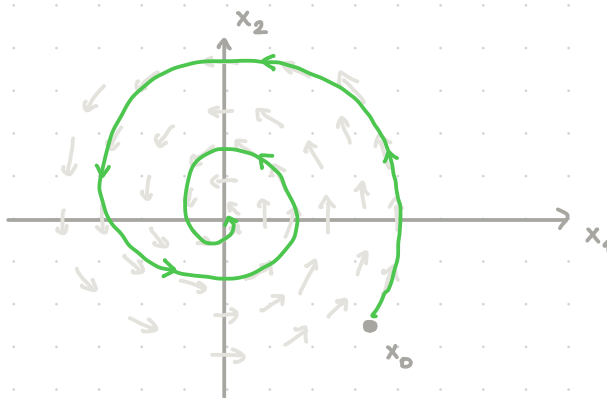
- Matrix exponentials?! 🤖🤖🤖🤖🤖🤖🤖🤖

⇒ It is actually really intuitive:

- In a first order system $x_{ic}(t) = e^{a \cdot t} \cdot x_0$ told us how our system reacted to an IC:



- A higher order $x_{ic} = e^{A \cdot t} \cdot x_0$ also shows exactly this:



Phase portrait of $\dot{x} = Ax$

- The matrix exponential therefore allows us to calculate the trajectory of any initial state.
- $e^{A \cdot t}$ is a description of our system's autonomous dynamics.

- Calculating $e^{A \cdot t}$ is also something you know from Lin Alg. We want either a diagonal or Jordan-Form A .

→ Diagonal:
$$\exp \left(\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} t \right) = \begin{bmatrix} e^{\lambda_1 \cdot t} & 0 \\ 0 & e^{\lambda_2 \cdot t} \end{bmatrix}$$

→ Jordan:
$$\exp \left(\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t \right) = \begin{bmatrix} e^{\lambda \cdot t} & t \cdot e^{\lambda \cdot t} \\ 0 & e^{\lambda \cdot t} \end{bmatrix}$$

→ Nilpotent ($A^k = 0$):
$$\exp \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} t \right) = \mathbb{I} + At + \frac{1}{2} (At)^2 + \dots$$

○ If our A is not in these forms, we can do coordinate transformations:

$$\begin{cases} \dot{x}(t) = A \cdot x(t) + B \cdot u(t) \\ y(t) = C \cdot x(t) + D \cdot u(t) \\ x(0) = x_0 \end{cases} \Rightarrow \begin{cases} \dot{\tilde{x}}(t) = (T^{-1}AT) \tilde{x}(t) + (T^{-1}B) u(t) \\ y(t) = CT \tilde{x}(t) + D u(t) \\ \tilde{x}(0) = T^{-1}x_0 \end{cases}$$

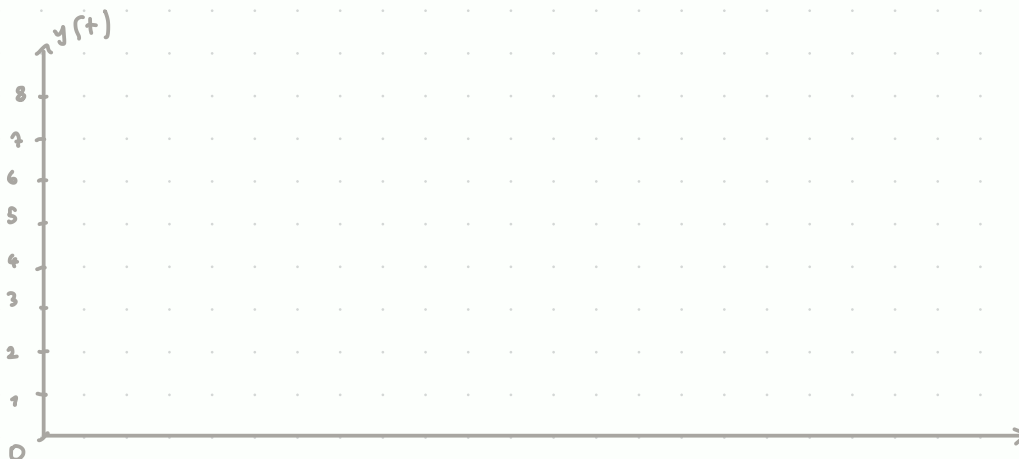
② Your turn:

Problem: Consider the following linear time-invariant system in state-space representation with state vector $x(t) \in \mathbb{R}^2$, input $u(t) \in \mathbb{R}$ and output $y(t) \in \mathbb{R}$.

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t) + 2u(t). \end{aligned}$$

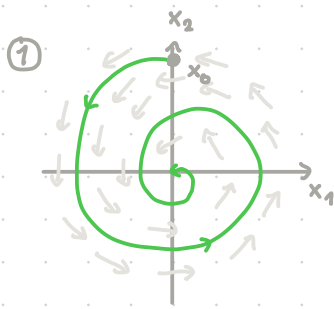
Q12 (1 Points) Calculate the initial condition response $y_{IC}(t)$ with the initial condition $x(0) = \begin{bmatrix} 1 & 3 \end{bmatrix}^T$.

Q13 (1 Points) Calculate the forced response $y_F(t)$, $t \geq 0$, to the (non-unit) step function $u(t) = 2$, $t \geq 0$.

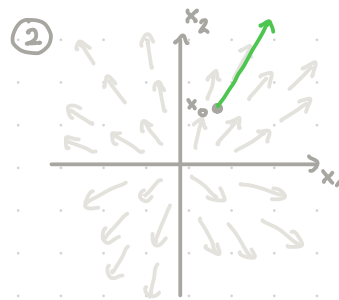


Stability

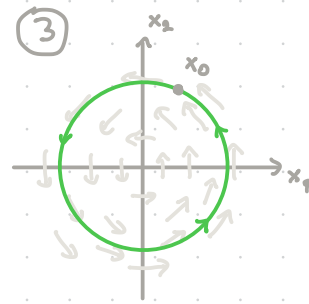
- We saw that higher order systems are naturally described by exponentials e^{At} . These exponentials can take many shapes:



In this case our x converges to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.



In this case our x diverges



In this case our x stays in a circle

- The critical piece of information we want to know for our systems is if they are **stable** or **not**.
- For the three systems above this can be decided visually. Let's give it a try.

①:

②:

③:

- We can also just look at the **eigenvalues** of A without having to draw anything, because each eigenvalue λ produces one term $e^{\lambda t}$ in $x(t)$.

→ λ with $\text{Re}(\lambda) < 0$: the term **decays** to 0.

→ λ with $\text{Re}(\lambda) > 0$: the term **diverges** to infinity.

→ λ with $\text{Re}(\lambda) = 0$: the term **neither grows nor decays**.

⚠ exception: a repeated $\text{Re}(\lambda) = 0$ eigenvalue in a Jordan block adds a $t \cdot e^{\lambda t}$ term which drifts away. $A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$

→ λ with $\text{Im}(\lambda) \neq 0$: the term **oscillates**.

- $x(t)$ is a sum of all these terms. One **diverging** term is enough to make the whole system diverge

$$x(t) = a \cdot e^{\lambda_1 t} + b \cdot e^{\lambda_2 t} + c \cdot e^{\lambda_3 t} \Rightarrow$$



⑦ Try to guess the eigenvalues of ①, ②, ③ ($>$, $<$ or $= 0$)

①:

②:

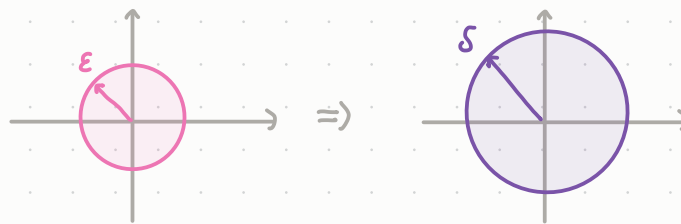
③:

- Now with these eigenvalues we can check for three formal definitions of stability.

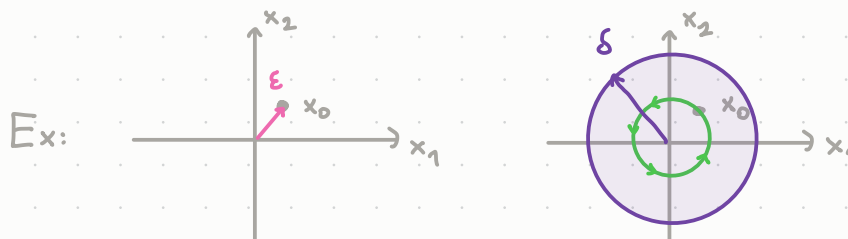
Lyapunov Stability:

- A system is Lyapunov stable if, for any bounded initial condition and zero input, the state remains bounded.

$$\|x_0\| < \varepsilon \quad \text{and} \quad u = 0 \quad \Rightarrow \quad \|x(t)\| < \delta \quad \forall t \geq 0$$



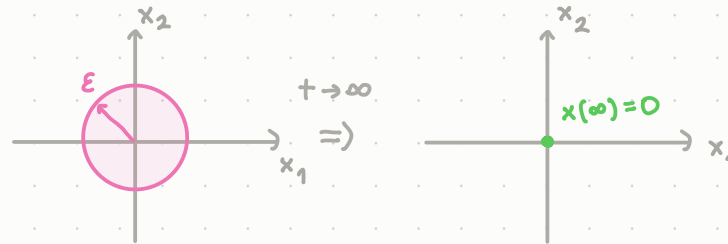
- True if $\text{Re}(\lambda) \leq 0 \quad \forall i$ and no Jordan-Block with $\text{Re}(\lambda) = 0$



Asymptotic Stability:

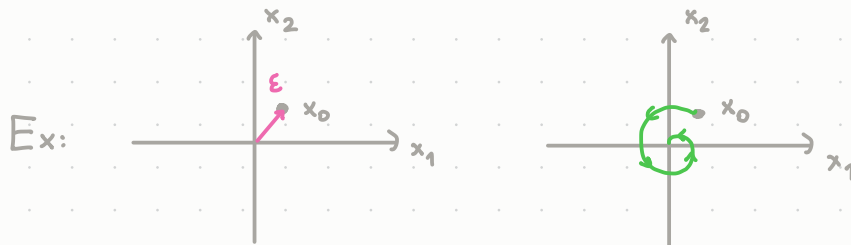
- A system is asymptotically stable if, for any bounded initial condition and zero input, the state converges to 0.

$$\|x_0\| < \varepsilon \quad \text{and} \quad u = 0 \quad \Rightarrow \quad \lim_{t \rightarrow \infty} \|x(t)\| = 0$$



- True if $\operatorname{Re}(\lambda) < 0 \quad \forall i$

- For linear systems: Asymptotic stability $\Rightarrow$ Lyapunov stability



BIBO Stability:

- A system is Bounded-Input, Bounded-Output stable if, for any bounded input, the output remains bounded

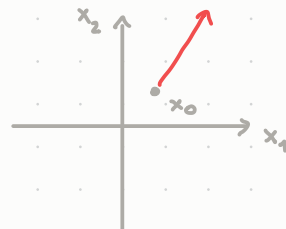
$$\|u(t)\| < \varepsilon \quad \forall t \geq 0, \quad \text{and} \quad x_0 = 0 \quad \Rightarrow \quad \|y(t)\| < \delta \quad \forall t \geq 0$$

- For linear systems: Asymptotic stability $\Rightarrow$ BIBO stability

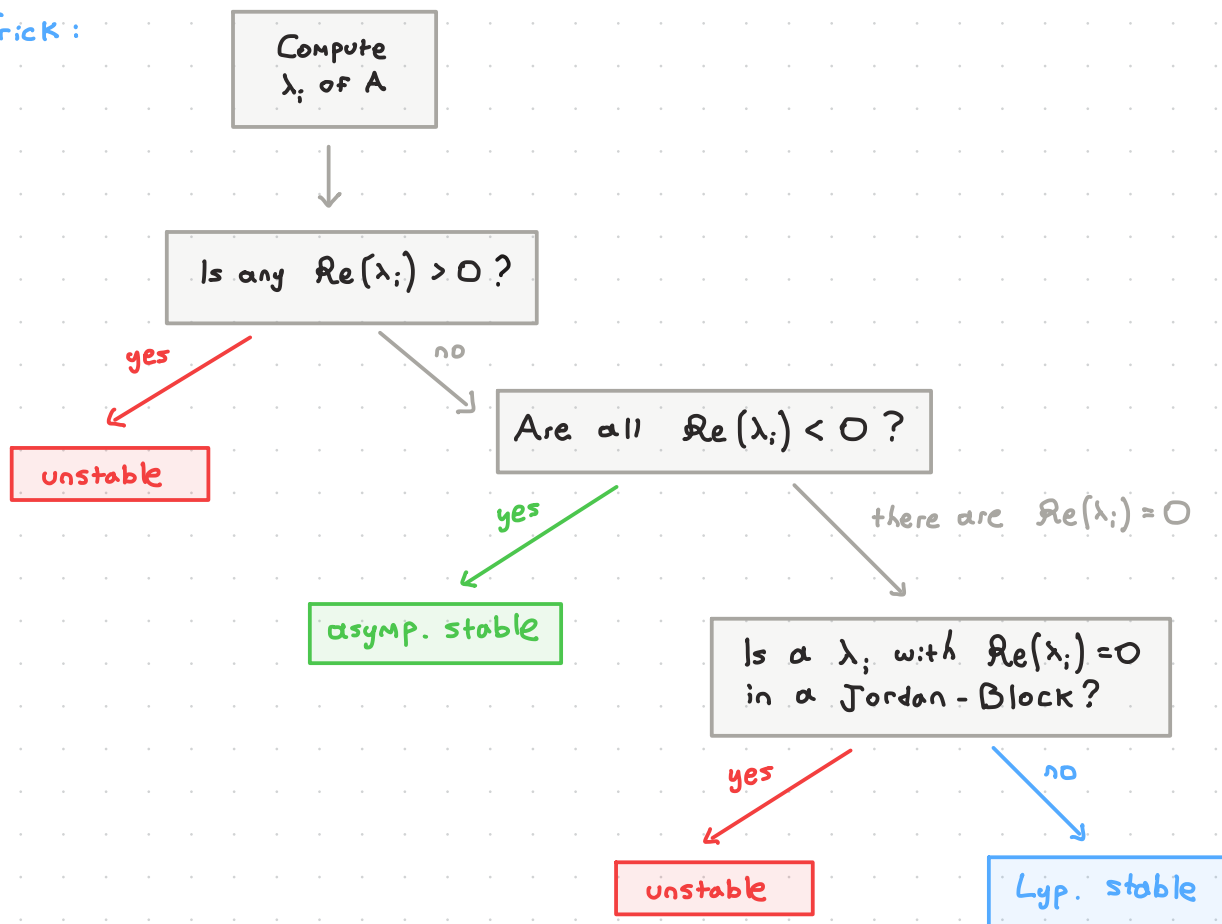
Unstable

- A system is unstable if it is not Lyapunov stable.

- So if $\operatorname{Re}(\lambda_i) > 0$
or a Jordan block with $\operatorname{Re}(\lambda) = 0$



Trick:



? Your Turn:

Problem: Consider the following linear time-invariant system in the state-space form given by:

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -2 & -1 \\ 1 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} x(t),\end{aligned}$$

where $x(t) \in \mathbb{R}^2$ represents the state of the system, $u(t) \in \mathbb{R}$ the input, and $y(t) \in \mathbb{R}$ the output.

Q12 (0.5 Points) Mark the correct answer.

The above system is:

- ☐ A Lyapunov stable but not asymptotically stable.
- ☐ B Asymptotically stable.
- ☐ C Unstable.

Problem: Consider the linear time-invariant system G in state-space form given by,

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix} x(t) + \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t),\end{aligned}$$

where $x(t)$ represents the state of the system, $u(t)$ the input and $y(t)$ the output.

Q11 (0.5 Points) Mark the correct answer.

The system G is,

- ☐ A asymptotically stable.
- ☐ B lyapunov stable.
- ☐ C unstable.

Problem: Consider the following linear time-invariant system in state-space form given by:

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -1 & \frac{1}{2} \\ 2 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ -1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} -1 & 1 \end{bmatrix} x(t) + 3 u(t),\end{aligned}$$

where $x(t)$ represents the state of the system, $u(t)$ the input and $y(t)$ the output.

Q11 (0.5 Points) Mark the correct answer.

The above system is:

- ☐ A lyapunov stable, but not asymptotically stable.
- ☐ B asymptotically stable.
- ☐ C unstable.